Roll No.....................

MID SEMESTER EXAMINATION 2025

Name of the Program: **B.Tech.**
Semester: **Third**

Name of the Course: **Discrete Structures and Combinatorics**
Course Code: **TMA 316**

Time: **90 Minutes**
Maximum Marks: **50**

Note:

(i) Answer **all the questions** by choosing ***any one of the sub questions***.

(ii) Each question carries 10 marks.

Q1	(10 marks)	
(a)	Let R be a binary relation defined as $$R = \{(a,b) \in R^2 : (a-b) \leq 3\}$$ Determine whether R is reflexive, symmetric, antisymmetric and transitive.	**CO1**
	OR	
(b)	Problem to the universal set U= {1,2,3,4,5,6,7,8,9} and the sets A= {1,2,3,4,5}, C= {5,6,7,8,9}, B= {4,5,6,7}, D= {2,4,6,8}, and E= {1,3,5,7,9} Find (i) (AUB) – (A∩B) (ii) (AUC) – (A∩D). (iii) (CUD) ∩ (CUB). (iv) A⊕D.	
Q2	(10 marks)	
(a)	If A is a set of positive integers and a relation R is defined on A as follows: $(a,b)R(c,d) \Leftrightarrow a+d = b+c \ \forall\, a,b,c,d \in R$, then prove that R is an equivalence relation.	**CO1**
	OR	
(b)	The function $f: R \to R$ defined as $f(x) = 2x+3,\ \forall\, x \in R$, and the function $g: I \to I$, defined as $g(x) = x^2+1, \forall\, x \in I$, Check whether composition gof is a bijection map or not.	
Q3	(10 marks)	
a)	If $\mathbf{D_m}$ denote the positive divisors of m ordered by divisibility. Draw the Hasse diagram for the following set and check which of them is Bounded and complemented lattices. **(i) $\mathbf{D_{30}}$** **(ii) $\mathbf{D_{12}}$**	**CO1**
	OR	
b)	In how many ways a 11 football player can be chosen out of 17 players when (i) five particular players are to be always include. (ii) two particular player are to be always excluded.	**CO3**
Q4	(10 marks)	
a)	The content of urns I, II III are as follows: In First urn contain 1 White, 2 Black and 3 Red balls, second urn contain 2 White, 1 Black and 1 Red balls and third Urn contain 4 White, 5 Black and 3 Red balls, one Urns is chosen at random and two balls drawn from it. They happen to be white and red. What is the probability that they come from Urns I?	**CO3**
	OR	

b)	Let X be a continuous random variable with probability density function $$f(x)=\begin{cases} k\,x & ;0\le x\le 1 \\ k & ;1\le x\le 2 \\ -kx+3k & ;2\le x\le 3 \\ 0 & ;elsewhere \end{cases}$$ (i)Determine the value of k (ii) Mean and variance of X.	
Q5	(10 marks)	
a)	The meaning and variance of a binomial variate X with parameters n and p are 16 and 8. Find the following: (i) $P(X=0)$ (ii) $P(X=1)$ (iii) $P(X\ge 2)$.	**CO3**
	OR	
b)	If X is a Poisson variate such that $P(X = 2) = 9P(X = 4) + 90P(X = 6)$. Find λ (the mean for X).	